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# =====
# MACHINE LEARNING PRACTICAL - MODULE I
# Data Preprocessing and Feature Engineering
# =====

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.feature_selection import SelectKBest, f_classif

# -----
# STUDENT DETAILS
# -----

print("=" * 70)
print("MACHINE LEARNING PRACTICAL - MODULE I")
print("Data Preprocessing and Feature Engineering")
print("=" * 70)

print("\nStudent Name : Pragati Parmar")
print("Roll Number   : BETN1AI25100")
print("Program        : B.Tech CSE (AI & ML)")
print("Subject         : Machine Learning")
print("Dataset         : Adult Income Dataset")

# -----
# DATASET URL
# -----

DATA_URL = "https://raw.githubusercontent.com/Pragati200706/Machine-Learning-Practical/main/adult.csv"

# -----
# Q1 - LOAD AND INSPECT DATASET
# -----

print("\n" + "=" * 70)
print("Q1 - DATASET LOADING AND INSPECTION")
print("=" * 70)

df = pd.read_csv(DATA_URL)

print("\nDataset Shape:")
print(df.shape)

print("\nFirst 10 Records:")

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display(df.head(10))

print("\nLast 10 Records:")
display(df.tail(10))

print("\nDataset Information:")
df.info()

print("\nDescriptive Statistics:")
display(df.describe(include="all").T)

print("\nStandard Missing Values:")
display(df.isnull().sum())

print("\nMissing Values Represented by '?':")
question_missing = (df == "?").sum()
display(question_missing[question_missing > 0])

# -----
# Q2 - DATA PREPROCESSING
# -----

print("\n" + "=" * 70)
print("Q2 - DATA PREPROCESSING")
print("=" * 70)

# Replace '?' with NaN
clean_df = df.replace("?", np.nan).copy()

print("\nMissing values before cleaning:")
print(clean_df.isnull().sum().sum())

# Fill missing values
for column in clean_df.columns:

    if clean_df[column].dtype == "object":
        clean_df[column] = clean_df[column].fillna(
            clean_df[column].mode()[0]
        )
    else:
        clean_df[column] = clean_df[column].fillna(
            clean_df[column].median()
        )

print("\nMissing values after cleaning:")
print(clean_df.isnull().sum().sum())

# Separate features and target
X = clean_df.drop("income", axis=1)
y = clean_df["income"]

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print("\nOriginal Feature Shape:")
print(X.shape)

# One-hot encode categorical variables
X = pd.get_dummies(X, drop_first=True)

print("\nFeature Shape After One-Hot Encoding:")
print(X.shape)

# Convert boolean columns to integers
X = X.astype(float)

# Feature scaling
scaler = StandardScaler()

X_scaled = pd.DataFrame(
    scaler.fit_transform(X),
    columns=X.columns
)

print("\nScaled Feature Sample:")
display(X_scaled.head())

# Encode target variable
y = y.map({
    "<=50K": 0,
    ">50K": 1
})

print("\nTarget Encoding:")
print("<=50K  -> 0")
print(">50K   -> 1")

print("\nEncoded Target Sample:")
display(y.head(10))

# -----
# Q3 - TRAIN / VALIDATION / TEST SPLIT
# -----

print("\n" + "=" * 70)
print("Q3 - TRAIN / VALIDATION / TEST SPLIT")
print("=" * 70)

# First split: 70% training, 30% temporary
X_train, X_temp, y_train, y_temp = train_test_split(
    X_scaled,
    y,
    test_size=0.30,
    random_state=42

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        stratify=y
    )

# Second split: temporary data into 15% validation and 15% testing
X_val, X_test, y_val, y_test = train_test_split(
    X_temp,
    y_temp,
    test_size=0.50,
    random_state=42,
    stratify=y_temp
)

print("\nDataset Partition:")
print("-" * 45)

print("Training Set   :", X_train.shape)
print("Validation Set :", X_val.shape)
print("Testing Set    :", X_test.shape)

print("\nNumber of Records:")
print("Training   :", len(X_train))
print("Validation :", len(X_val))
print("Testing    :", len(X_test))

# -----
# Q4 - SELECTKBEST FEATURE SELECTION
# -----

print("\n" + "=" * 70)
print("Q4 - SELECTKBEST FEATURE SELECTION")
print("=" * 70)

selector = SelectKBest(
    score_func=f_classif,
    k=10
)

selector.fit(
    X_train,
    y_train
)

selected_features = X_train.columns[
    selector.get_support()
]

feature_scores = pd.DataFrame({
    "Feature": X_train.columns,
    "F-Score": selector.scores_
})

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feature_scores = feature_scores.sort_values(
    by="F-Score",
    ascending=False
).reset_index(drop=True)

print("\nTop 10 Selected Features:")

for i, feature in enumerate(selected_features, start=1):
    print(f"{i:2}. {feature}")

print("\nTop 10 Feature Scores:")
display(feature_scores.head(10))

# -----
# Q5 - VISUALIZATION
# -----

print("\n" + "=" * 70)
print("Q5 - DATA VISUALIZATION")
print("=" * 70)

# =====
# GRAPH 1 - INCOME DISTRIBUTION
# =====

plt.figure(figsize=(8, 5))

income_counts = df["income"].value_counts()

bars = plt.bar(
    income_counts.index.astype(str),
    income_counts.values
)

plt.title(
    "Income Distribution Before Preprocessing",
    fontsize=14,
    pad=12
)

plt.xlabel(
    "Income Category",
    fontsize=11
)

plt.ylabel(
    "Number of Records",
    fontsize=11

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)

plt.xticks(
    rotation=0,
    fontsize=10
)

# Add values above bars
for bar in bars:
    height = bar.get_height()

    plt.text(
        bar.get_x() + bar.get_width() / 2,
        height,
        f"{int(height):,}",
        ha="center",
        va="bottom",
        fontsize=10
    )

plt.tight_layout()
plt.show()

# =====
# GRAPH 2 - MISSING VALUES
# =====

missing = (df == "?").sum()
missing = missing[missing > 0]

plt.figure(figsize=(9, 5))

bars = plt.bar(
    missing.index.astype(str),
    missing.values
)

plt.title(
    "Missing Values Before Preprocessing",
    fontsize=14,
    pad=12
)

plt.xlabel(
    "Features",
    fontsize=11
)

plt.ylabel(
    "Number of Missing Values",
    fontsize=11

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        fontsize=11
    )

    plt.xticks(
        rotation=30,
        ha="right",
        fontsize=9
    )

    for bar in bars:
        height = bar.get_height()

        plt.text(
            bar.get_x() + bar.get_width() / 2,
            height,
            f"{int(height):,}",
            ha="center",
            va="bottom",
            fontsize=8
        )

    plt.tight_layout()
    plt.show()

# =====
# GRAPH 3 - TOP 10 FEATURE SCORES
# =====

top_features = feature_scores.head(10).sort_values(
    by="F-Score",
    ascending=True
)

plt.figure(figsize=(9, 6))

plt.barh(
    top_features["Feature"].astype(str),
    top_features["F-Score"]
)

plt.title(
    "Top 10 Features Selected Using SelectKBest",
    fontsize=14,
    pad=12
)

plt.xlabel(
    "F-Score",
    fontsize=11
)

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plt.ylabel(
    "Feature",
    fontsize=11
)

plt.tight_layout()
plt.show()

# =====
# GRAPH 4 - CORRELATION MATRIX
# =====

# Select numerical features with manageable dimensions
correlation_matrix = X_scaled.corr()

plt.figure(figsize=(10, 8))

sns.heatmap(
    correlation_matrix,
    cmap="coolwarm",
    annot=False,
    linewidths=0.1,
    cbar=True
)

plt.title(
    "Correlation Matrix After Preprocessing",
    fontsize=14,
    pad=12
)

plt.xticks(
    rotation=45,
    ha="right",
    fontsize=7
)

plt.yticks(
    rotation=0,
    fontsize=7
)

plt.tight_layout()
plt.show()

# -----
# BEFORE VS AFTER SUMMARY
# -----

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print("\n" + "=" * 70)
print("BEFORE VS AFTER PREPROCESSING SUMMARY")
print("=" * 70)

original_rows, original_columns = df.shape
processed_rows, processed_columns = X_scaled.shape

print("\nOriginal Dataset:")
print("Rows      :", original_rows)
print("Columns :", original_columns)

print("\nProcessed Feature Dataset:")
print("Rows      :", processed_rows)
print("Columns :", processed_columns)

print("\nMissing Values Before Cleaning:")
print(df.isnull().sum().sum() + (df == "?").sum().sum())

print("\nMissing Values After Cleaning:")
print(X_scaled.isnull().sum().sum())

print("\nCategorical Features:")
print("Encoded using One-Hot Encoding")

print("\Numerical Features:")
print("Scaled using StandardScaler")

print("\nFeature Selection:")
print("SelectKBest with ANOVA F-test")

print("\nData Split:")
print("70% Training / 15% Validation / 15% Testing")

# -----
# SAVE PREPROCESSED DATASET
# -----

processed_dataset = X_scaled.copy()
processed_dataset["income"] = y.values

processed_dataset.to_csv(
    "adult_preprocessed_pragati.csv",
    index=False
)

print("\nProcessed dataset saved as:")
print("adult_preprocessed_pragati.csv")

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# FINAL CONCLUSION
# -----

print("\n" + "=" * 70)
print("MODULE I COMPLETED SUCCESSFULLY")
print("=" * 70)

print("""
Conclusion:
The Adult Income dataset was inspected and preprocessed successfully.
Missing values represented by '?' were handled, categorical variables
were converted using one-hot encoding, numerical features were scaled,
and the dataset was divided into training, validation and testing sets.
SelectKBest with ANOVA F-test was used for feature selection.
The preprocessing results and selected features were visualized using
appropriate charts and a correlation matrix.
""")

print("=" * 70)
```


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MACHINE LEARNING PRACTICAL - MODULE I

Data Preprocessing and Feature Engineering

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Student Name : Pragati Parmar

Roll Number : BETN1AI25100

Program : B.Tech CSE (AI & ML)

Subject : Machine Learning

Dataset : Adult Income Dataset

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Q1 - DATASET LOADING AND INSPECTION

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Dataset Shape:

(32561, 15)

First 10 Records:

	age	workclass	fnlwgt	education	education.num	marital.status	occupation	relationship	race	sex	capital.gain	capital.loss	hours.per.week	native.country
0	90	?	77053	HS-grad	9	Widowed	?	Not-in-family	White	Female	0	4356	40	United-States
1	82	Private	132870	HS-grad	9	Widowed	Exec-managerial	Not-in-family	White	Female	0	4356	18	United-States
2	66	?	186061	Some-college	10	Widowed	?	Unmarried	Black	Female	0	4356	40	United-States
3	54	Private	140359	7th-8th	4	Divorced	Machine-op-inspct	Unmarried	White	Female	0	3900	40	United-States
4	41	Private	264663	Some-college	10	Separated	Prof-specialty	Own-child	White	Female	0	3900	40	United-States
5	34	Private	216864	HS-grad	9	Divorced	Other-service	Unmarried	White	Female	0	3770	45	United-States
6	38	Private	150601	10th	6	Separated	Adm-clerical	Unmarried	White	Male	0	3770	40	United-States
7	74	State-gov	88638	Doctorate	16	Never-married	Prof-specialty	Other-relative	White	Female	0	3683	20	United-States
8	68	Federal-gov	422013	HS-grad	9	Divorced	Prof-specialty	Not-in-family	White	Female	0	3683	40	United-States
9	41	Private	70037	Some-college	10	Never-married	Craft-repair	Unmarried	White	Male	0	3004	60	?

Last 10 Records:

	age	workclass	fnlwgt	education	education.num	marital.status	occupation	relationship	race	sex	capital.gain	capital.loss	hours.per.week	native.country
32551	43	Self-emp-not-inc	27242	Some-college	10	Married-civ-spouse	Craft-repair	Husband	White	Male	0	0	50	United-S
...	...	...	...	...	...	...	...	...	Amer-	...	...	...	...	...

32552	32	Private	34066	10th	6	Married-civ-spouse	Handlers-cleaners	Husband	Indian-Eskimo	Male	0	0	40	United-S
32553	43	Private	84661	Assoc-voc	11	Married-civ-spouse	Sales	Husband	White	Male	0	0	45	United-S
32554	32	Private	116138	Masters	14	Never-married	Tech-support	Not-in-family	Asian-Pac-Islander	Male	0	0	11	Te
32555	53	Private	321865	Masters	14	Married-civ-spouse	Exec-managerial	Husband	White	Male	0	0	40	United-S
32556	22	Private	310152	Some-college	10	Never-married	Protective-serv	Not-in-family	White	Male	0	0	40	United-S
32557	27	Private	257302	Assoc-acdm	12	Married-civ-spouse	Tech-support	Wife	White	Female	0	0	38	United-S
32558	40	Private	154374	HS-grad	9	Married-civ-spouse	Machine-op-inspct	Husband	White	Male	0	0	40	United-S
32559	58	Private	151910	HS-grad	9	Widowed	Adm-clerical	Unmarried	White	Female	0	0	40	United-S
32560	22	Private	201490	HS-grad	9	Never-married	Adm-clerical	Own-child	White	Male	0	0	20	United-S

Dataset Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 32561 entries, 0 to 32560

Data columns (total 15 columns):

#	Column	Non-Null Count	Dtype
0	age	32561 non-null	int64
1	workclass	32561 non-null	object
2	fnlwgt	32561 non-null	int64
3	education	32561 non-null	object
4	education.num	32561 non-null	int64
5	marital.status	32561 non-null	object
6	occupation	32561 non-null	object
7	relationship	32561 non-null	object
8	race	32561 non-null	object
9	sex	32561 non-null	object
10	capital.gain	32561 non-null	int64
11	capital.loss	32561 non-null	int64
12	hours.per.week	32561 non-null	int64
13	native.country	32561 non-null	object
14	income	32561 non-null	object

dtypes: int64(6), object(9)

memory usage: 3.7+ MB

Descriptive Statistics:

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
age	32561.0	NaN	NaN	NaN	38.581647	13.640433	17.0	28.0	37.0	48.0	90.0

workclass	32561	9	Private	22696	NaN	NaN	NaN	NaN	NaN	NaN	NaN
fnlwgt	32561.0	NaN	NaN	NaN	189778.366512	105549.977697	12285.0	117827.0	178356.0	237051.0	1484705.0
education	32561	16	HS-grad	10501	NaN	NaN	NaN	NaN	NaN	NaN	NaN
education.num	32561.0	NaN	NaN	NaN	10.080679	2.57272	1.0	9.0	10.0	12.0	16.0
marital.status	32561	7	Married-civ-spouse	14976	NaN	NaN	NaN	NaN	NaN	NaN	NaN
occupation	32561	15	Prof-specialty	4140	NaN	NaN	NaN	NaN	NaN	NaN	NaN
relationship	32561	6	Husband	13193	NaN	NaN	NaN	NaN	NaN	NaN	NaN
race	32561	5	White	27816	NaN	NaN	NaN	NaN	NaN	NaN	NaN
sex	32561	2	Male	21790	NaN	NaN	NaN	NaN	NaN	NaN	NaN
capital.gain	32561.0	NaN	NaN	NaN	1077.648844	7385.292085	0.0	0.0	0.0	0.0	99999.0
capital.loss	32561.0	NaN	NaN	NaN	87.30383	402.960219	0.0	0.0	0.0	0.0	4356.0
hours.per.week	32561.0	NaN	NaN	NaN	40.437456	12.347429	1.0	40.0	40.0	45.0	99.0
native.country	32561	42	United-States	29170	NaN	NaN	NaN	NaN	NaN	NaN	NaN
income	32561	2	<=50K	24720	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Standard Missing Values:

	0
age	0
workclass	0
fnlwgt	0
education	0
education.num	0
marital.status	0
occupation	0
relationship	0
race	0
sex	0
capital.gain	0
capital.loss	0
hours.per.week	0
native.country	0
income	0

dtype: int64

Missing Values Represented by '?':

	0
workclass	1836
occupation	1843
native.country	583

dtype: int64

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Q2 - DATA PREPROCESSING

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Missing values before cleaning:

4262

Missing values after cleaning:

0

Original Feature Shape:

(32561, 14)

Feature Shape After One-Hot Encoding:

(32561, 97)

Scaled Feature Sample:

	age	fnlwgt	education.num	capital.gain	capital.loss	hours.per.week	workclass_Local- gov	workclass_Never- worked	workclass_Private	workclass_Self- emp-inc	...	native.cou
0	3.769612	-1.067997	-0.420060	-0.14592	10.593507	-0.035429	-0.262097	-0.014664	0.57209	-0.188389	...	
1	3.183112	-0.539169	-0.420060	-0.14592	10.593507	-1.817204	-0.262097	-0.014664	0.57209	-0.188389	...	
2	2.010110	-0.035220	-0.031360	-0.14592	10.593507	-0.035429	-0.262097	-0.014664	0.57209	-0.188389	...	
3	1.130359	-0.468215	-2.363558	-0.14592	9.461864	-0.035429	-0.262097	-0.014664	0.57209	-0.188389	...	
4	0.177296	0.709482	-0.031360	-0.14592	9.461864	-0.035429	-0.262097	-0.014664	0.57209	-0.188389	...	

5 rows × 97 columns

Target Encoding:

<=50K -> 0

>50K -> 1

Encoded Target Sample:

	income
0	0
1	0

```
2      0
3      0
4      0
5      0
6      0
7      1
8      0
9      1
```

dtype: int64

```
=====
Q3 - TRAIN / VALIDATION / TEST SPLIT
=====
```

Dataset Partition:

```
-----
Training Set   : (22792, 97)
Validation Set : (4884, 97)
Testing Set    : (4885, 97)
```

Number of Records:

```
Training   : 22792
Validation : 4884
Testing    : 4885
```

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Q4 - SELECTKBEST FEATURE SELECTION
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Top 10 Selected Features:

1. age
2. education.num
3. capital.gain
4. hours.per.week
5. marital.status_Married-civ-spouse
6. marital.status_Never-married
7. occupation_Exec-managerial
8. relationship_Not-in-family
9. relationship_Own-child
10. sex_Male

Top 10 Feature Scores:

	Feature	F-Score
0	marital.status_Married-civ-spouse	5720.914988
1	education.num	2861.022847
2	marital.status_Never-married	2573.895611

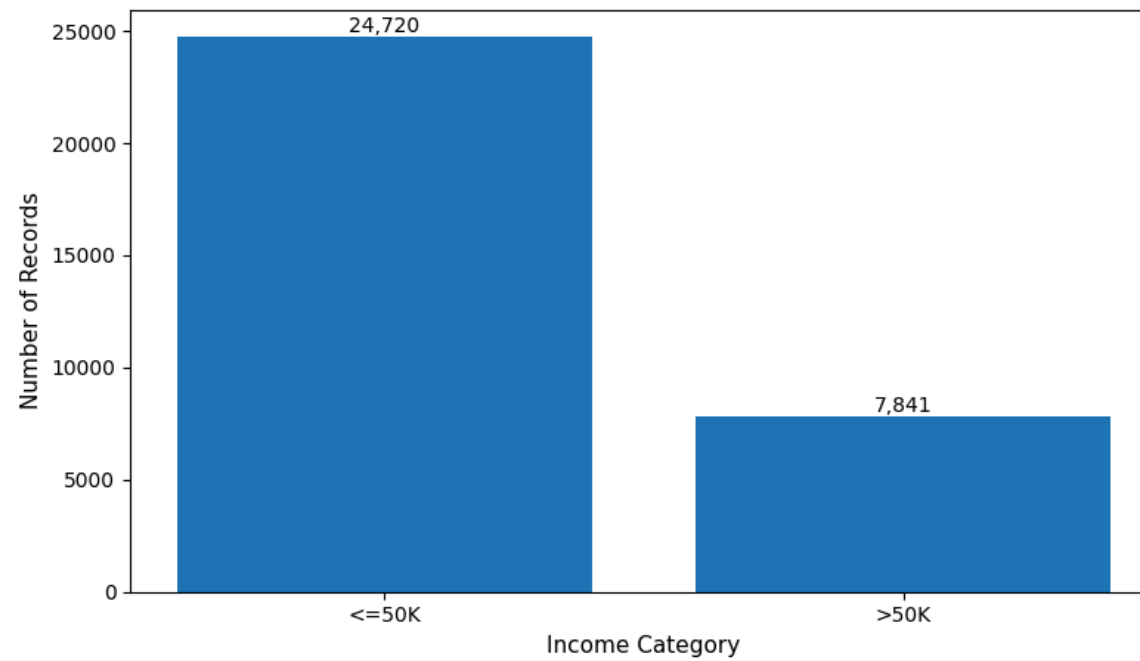
3	hours.per.week	1283.762720
4	relationship_Own-child	1280.865957
5	age	1265.034148
6	capital.gain	1216.572152
7	occupation_Exec-managerial	1119.510360
8	sex_Male	1114.230534
9	relationship_Not-in-family	821.762413

=====

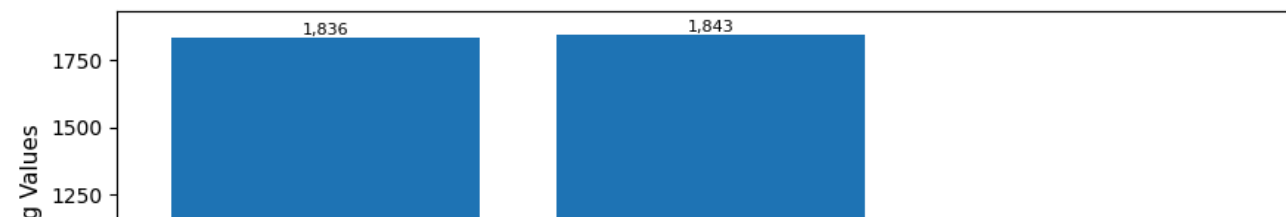
Q5 - DATA VISUALIZATION

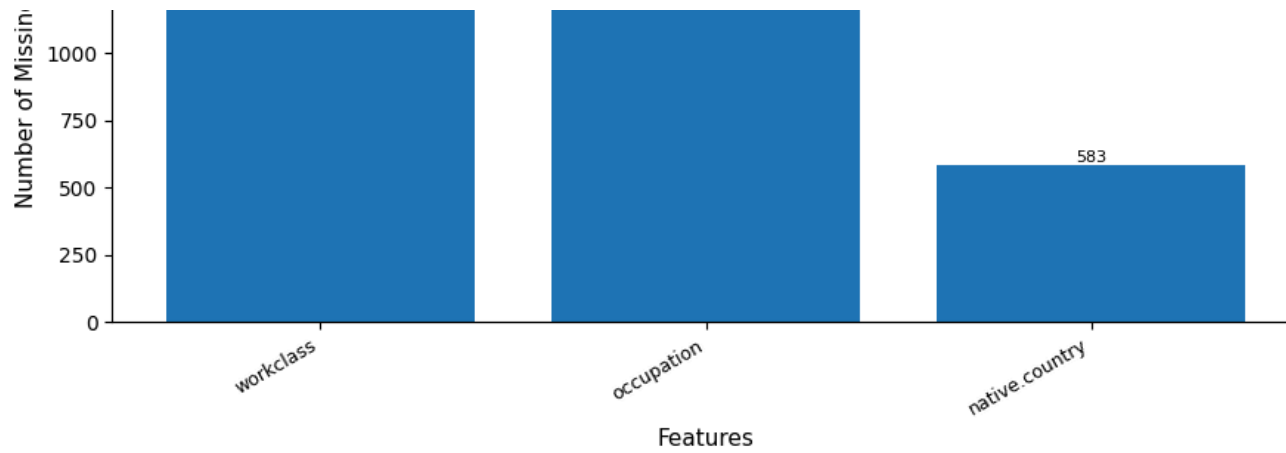
=====

Income Distribution Before Preprocessing

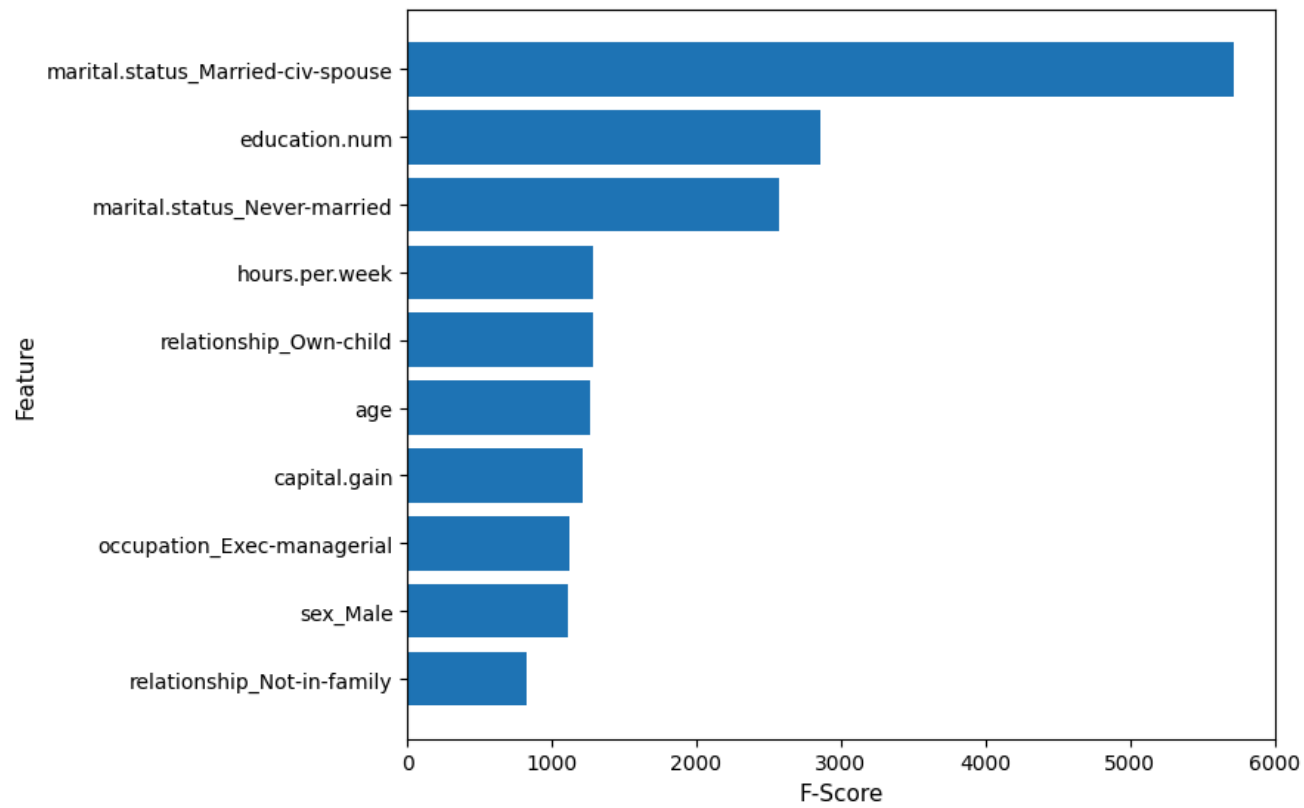


Missing Values Before Preprocessing





Top 10 Features Selected Using SelectKBest



Correlation Matrix After Preprocessing



